

Laboratory work 3

Tasks:

1. Generate and insert 200 random rows in your airport database.

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'Tables (17)' expanded. The main query editor shows an SQL script to insert 200 random rows into the 'passengers' table. The script uses the 'generate_series' function to create a sequence of IDs and random data for names, birth dates, genders, and passport numbers. The 'Data Output' tab at the bottom shows the first 7 rows of the inserted data.

```
1 INSERT INTO passengers (
2   passenger_id, first_name, last_name, date_of_birth, gender,
3   country_of_citizenship, country_of_residence,
4   passport_number, created_at, updated_at
5 )
6 SELECT
7   (SELECT COALESCE(MAX(passenger_id), 0) FROM passengers) + g,
8   (ARRAY['Ali', 'Dana', 'Aida', 'Timur', 'Mira'])[floor(random() * 5)::int + 1],
9   (ARRAY['Asanov', 'Omarov', 'Bekova', 'Saparova', 'Kim'])[floor(random() * 5)::int + 1],
10  DATE '1980-01-01' + floor(random() * 12000)::int,
11  (ARRAY['Male', 'Female'])[floor(random() * 2)::int + 1],
12  'Kazakhstan',
13  'Kazakhstan',
14  'LAB3-' || g || '-' || floor(random() * 1000000)::int,
15  CURRENT_TIMESTAMP,
16  CURRENT_TIMESTAMP
17 FROM generate_series(1, 200) AS g
18 RETURNING passenger_id, first_name, last_name, passport_number;
```

passenger_id	first_name	last_name	passport_number
1	Dana	Saparova	LAB3-1-215129
2	Dana	Bekova	LAB3-2-342548
3	Aida	Bekova	LAB3-3-807963
4	Ali	Omarov	LAB3-4-899238
5	Mira	Omarov	LAB3-5-118886
6	Timur	Saparova	LAB3-6-866784
7	Dana	Omarov	LAB3-7-688461

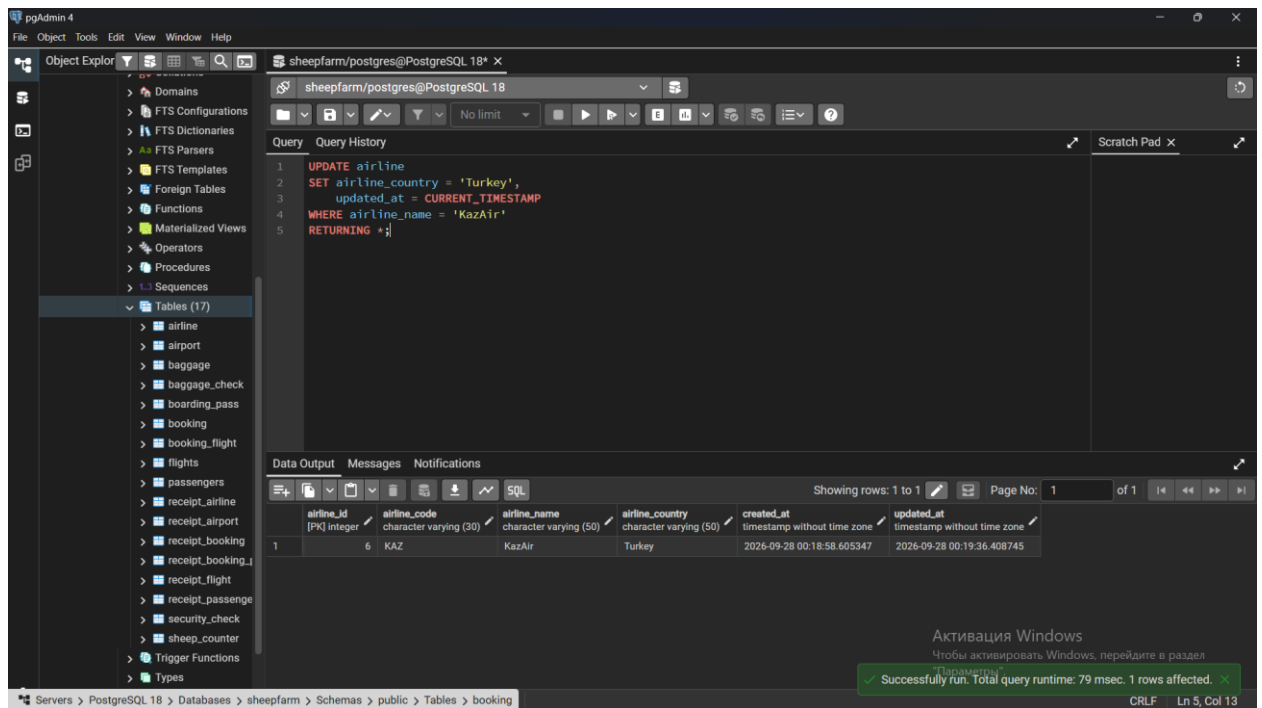
2. Add a new airline named "KazAir" based in "Kazakhstan" to the airline table.

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with 'Tables (17)' expanded. The main query editor shows an SQL script to insert a new airline named 'KazAir' into the 'airline' table. The script uses the 'generate_series' function to create a sequence of IDs and random data for names, birth dates, genders, and passport numbers. The 'Data Output' tab at the bottom shows the first 1 row of the inserted data.

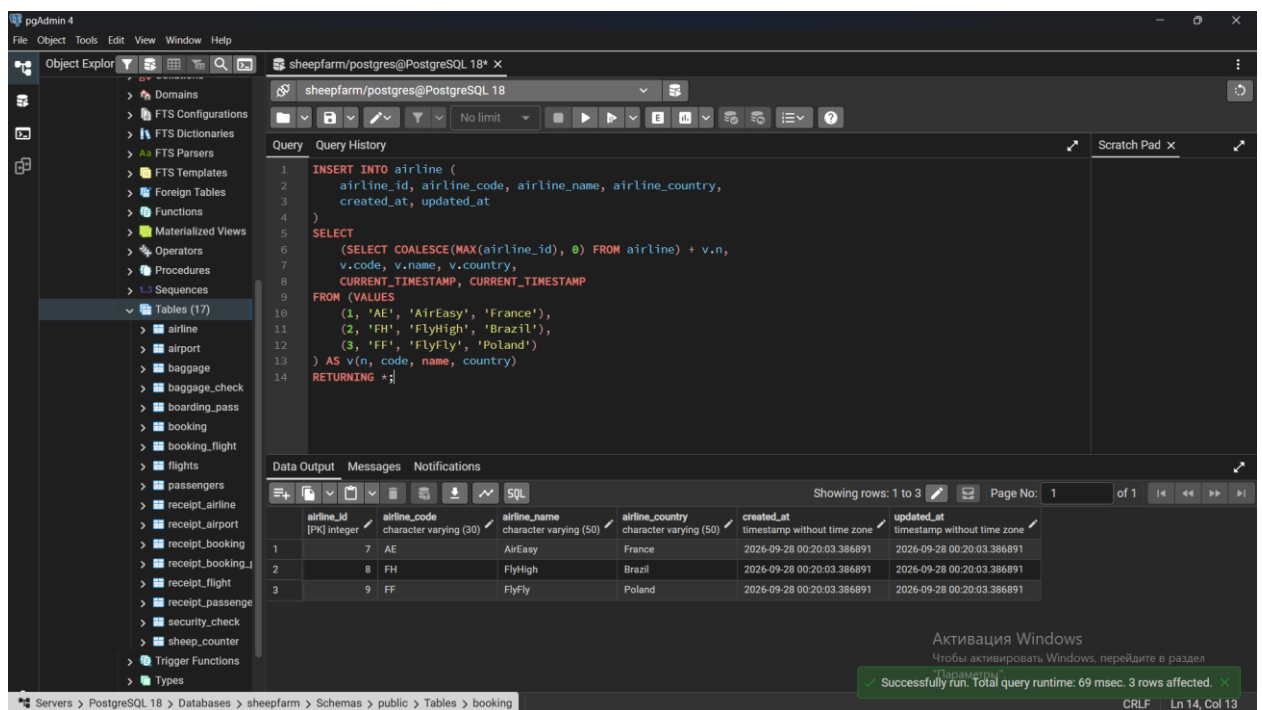
```
1 INSERT INTO airline (
2   airline_id, airline_code, airline_name, airline_country,
3   created_at, updated_at
4 )
5 VALUES (
6   (SELECT COALESCE(MAX(airline_id), 0) + 1 FROM airline),
7   'KAZ', 'KazAir', 'Kazakhstan',
8   CURRENT_TIMESTAMP, CURRENT_TIMESTAMP
9 )
10 RETURNING *;
```

airline_id	airline_code	airline_name	airline_country	created_at	updated_at
6	KAZ	KazAir	Kazakhstan	2026-09-28 00:18:58.605347	2026-09-28 00:18:58.605347

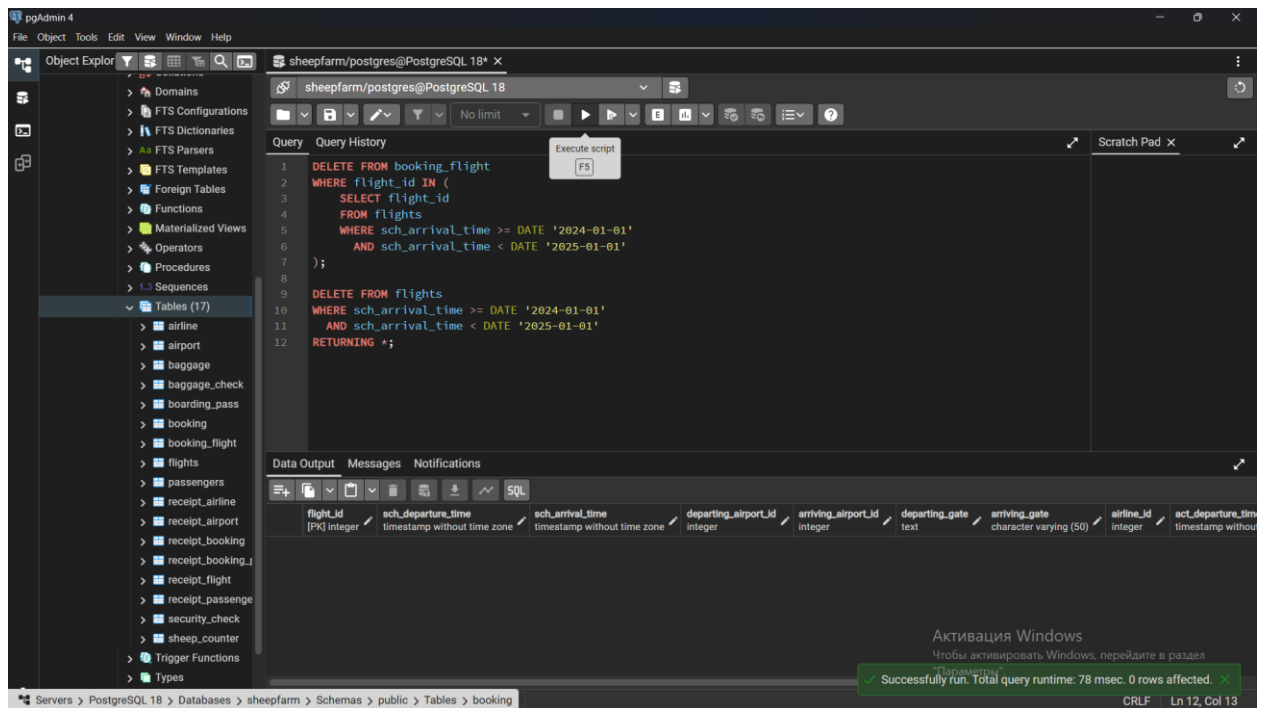
3. Update the airline country "KazAir" to "Turkey".



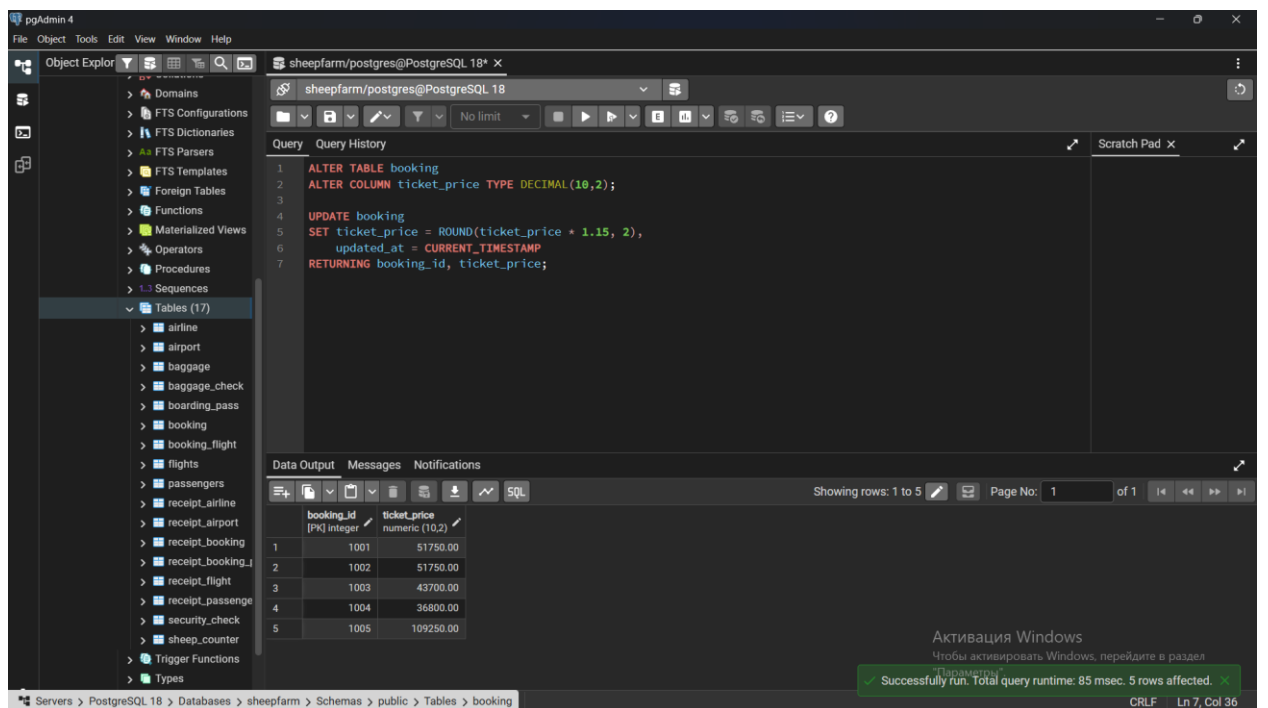
4. Add three airlines at once: "AirEasy" in "France", "FlyHigh" in "Brazil" and "FlyFly" in "Poland".



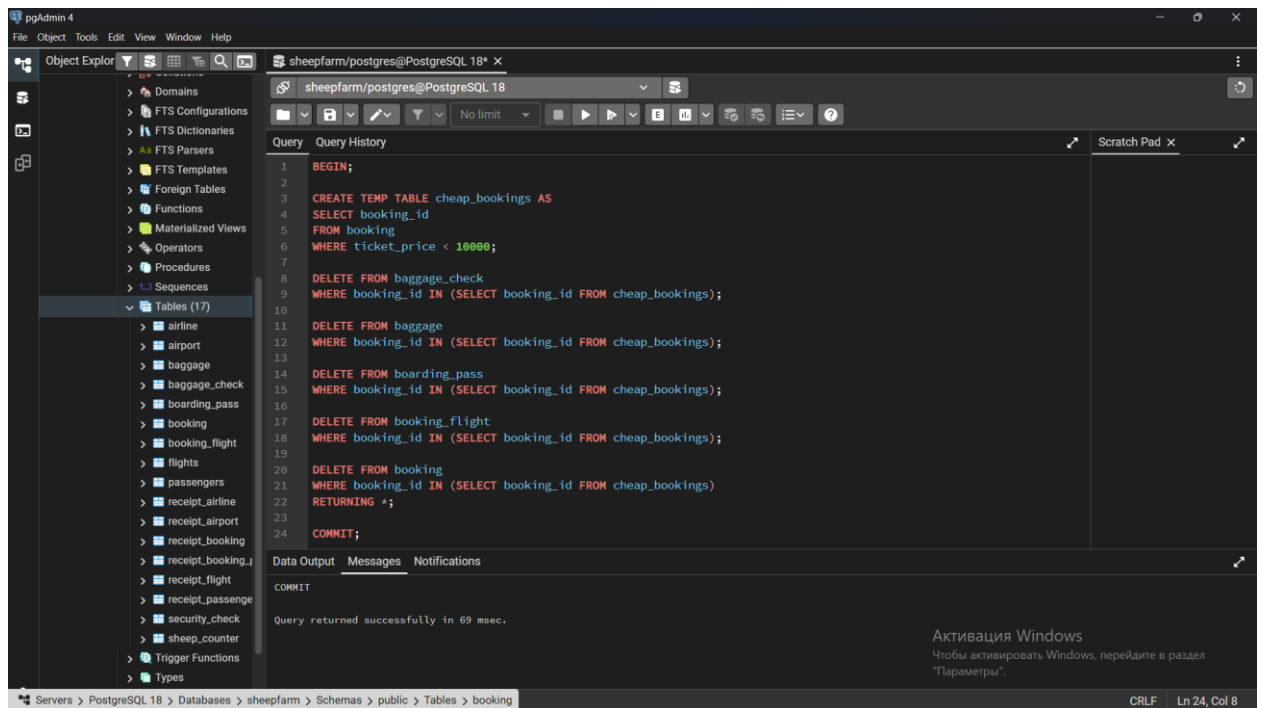
5. Delete all flights whose arrival in 2024 year.



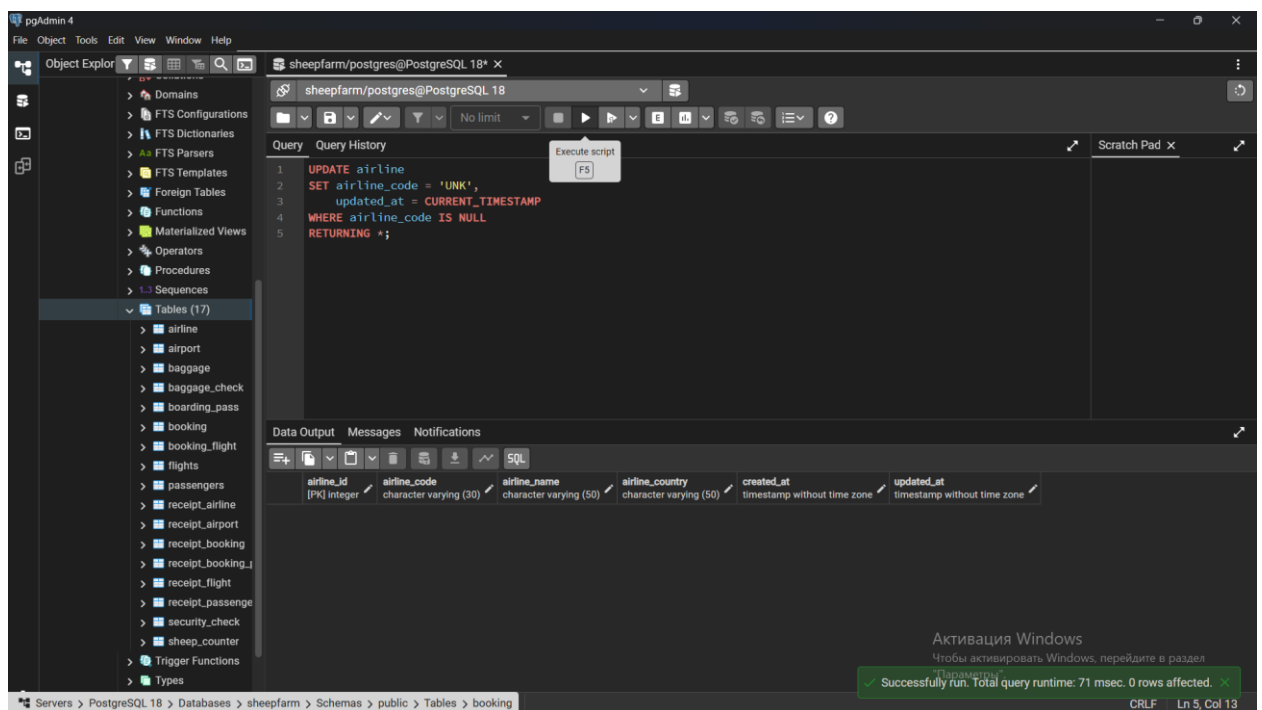
6. Increase the price of all tickets in booking table for flights by 15%.



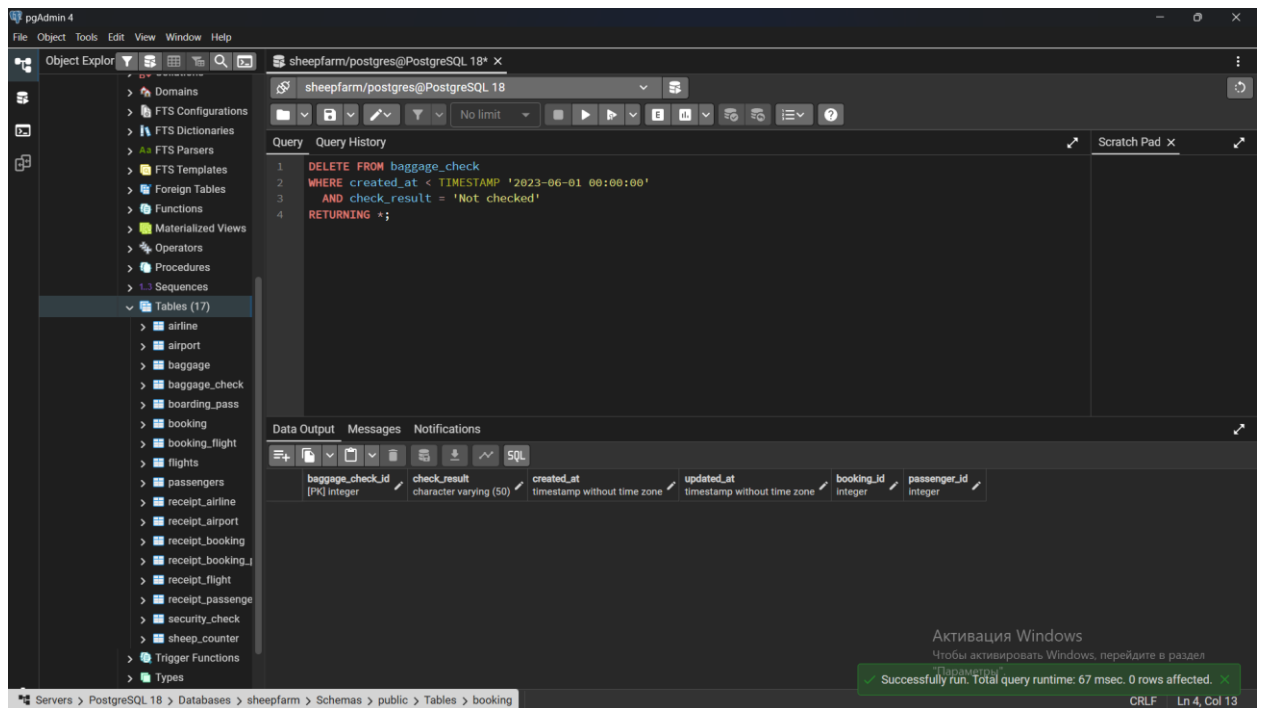
7. Delete all tickets where price is less than 10000.



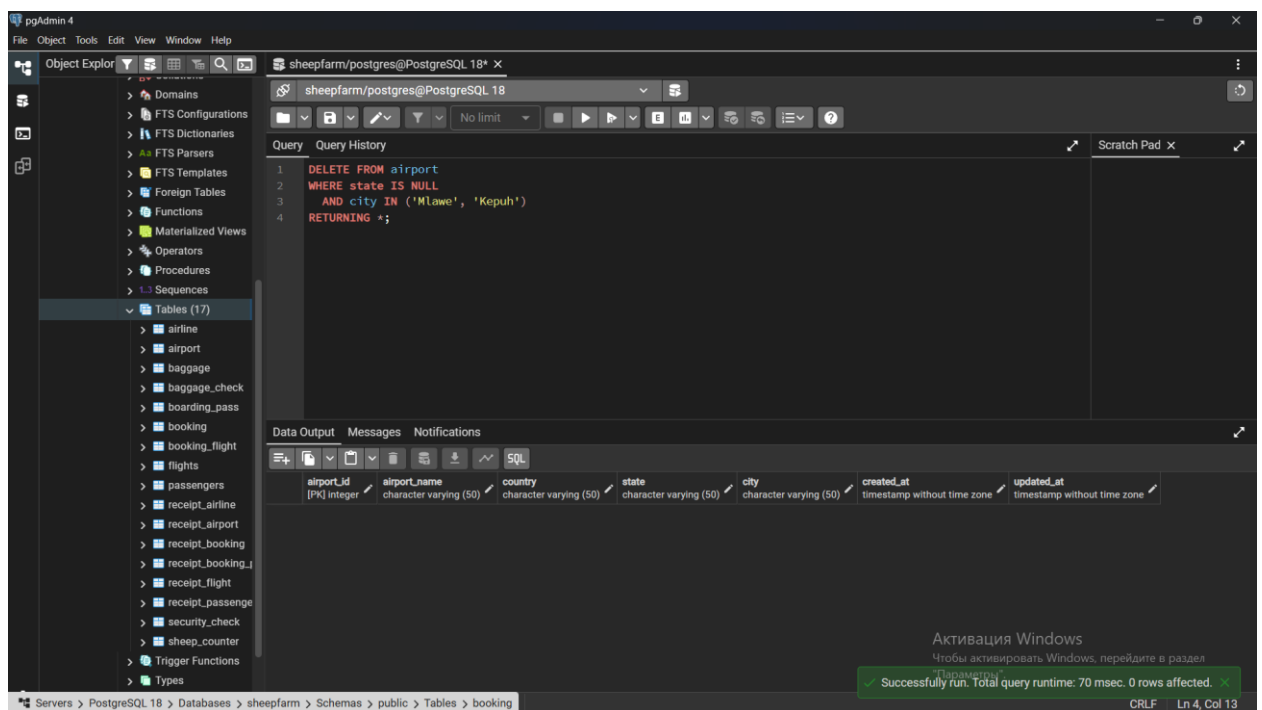
8. Set a default code 'UNK' (Unknown) for any airline in the airline table where airline_code is missing (NULL).



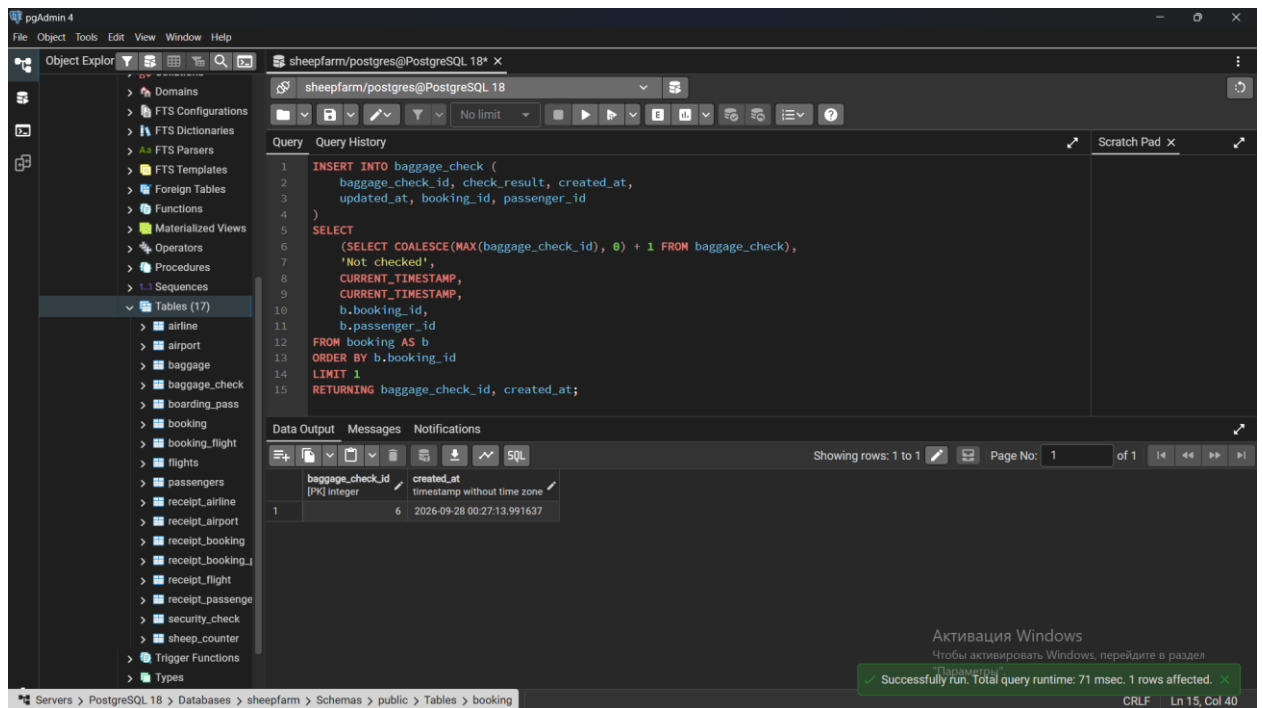
9. Delete all baggage check records created before June 1, 2023, whose result is still 'Not checked'.



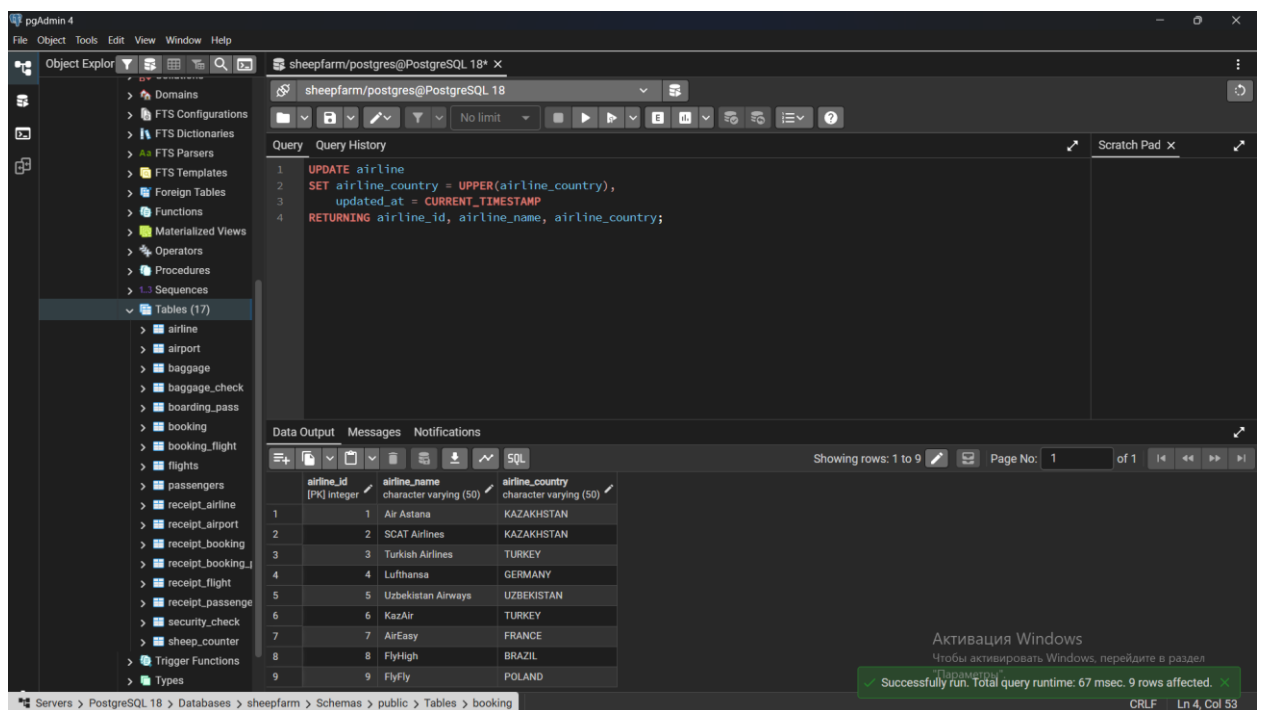
10. Delete all entries from the airport table where state is missing (NULL) and the city is either 'Mlawe' or 'Kepuh'.



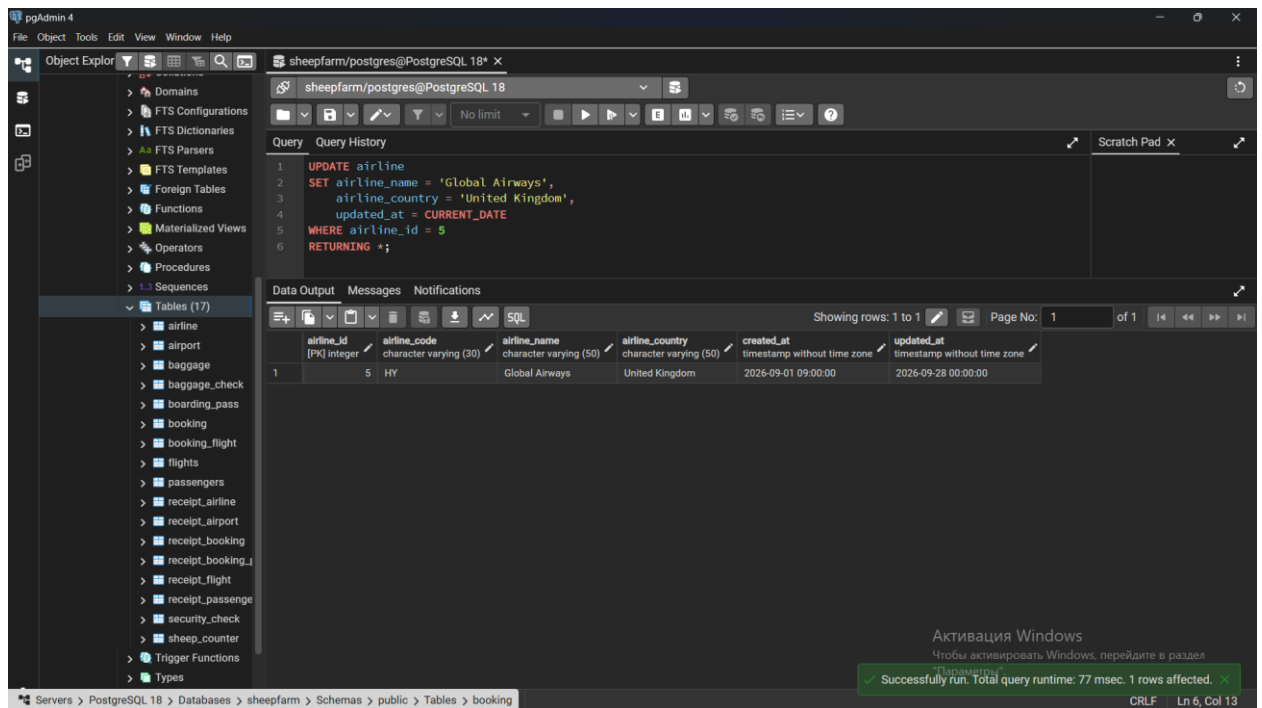
11. Insert a new unchecked baggage inspection record and immediately return its newly generated baggage_check_id and created_at timestamp.



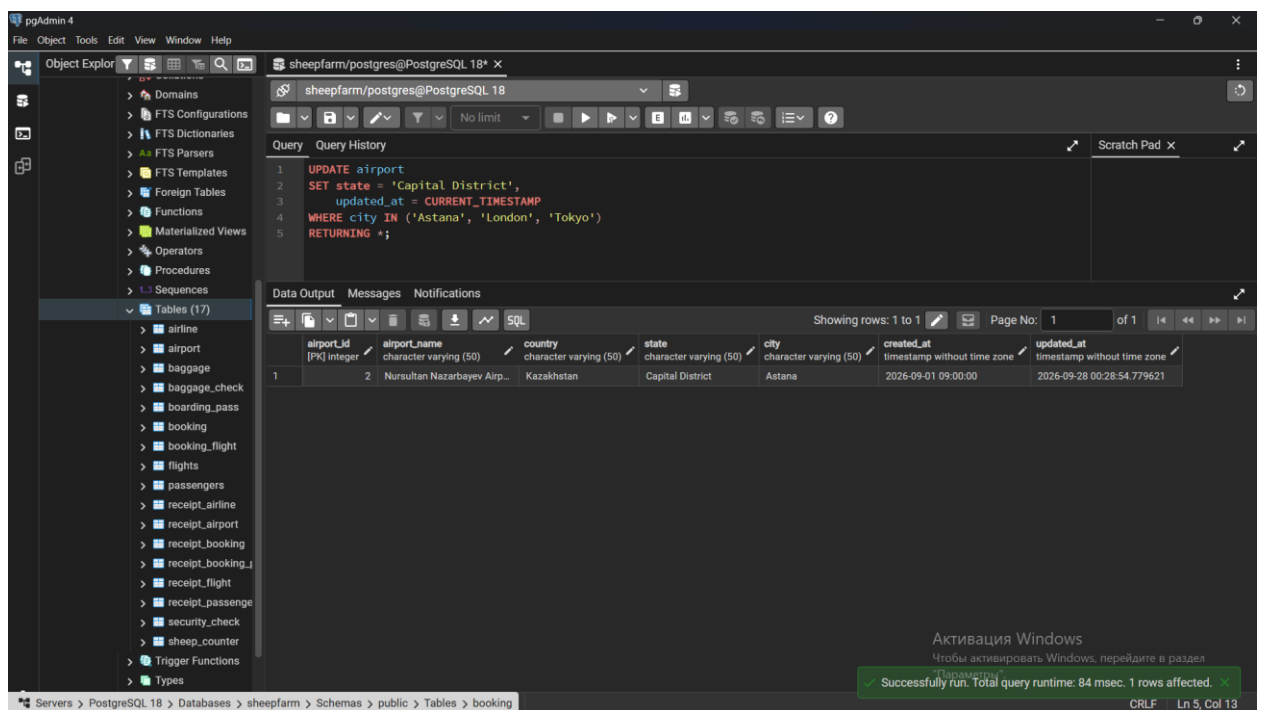
12. Convert all country names in the airline table to uppercase to ensure data consistency.



13. Update both the name and country for the airline with airline_id = 5. Set the name to 'Global Airways' and the country to 'United Kingdom', while updating update_at to the current date.



14. Set the state field to 'Capital District' for airports located in specific cities: 'Astana', 'London', and 'Tokyo'



15. Mark all baggage check records (baggage_check) created in March 2024 with a status of 'Not checked' as completed ('Checked').

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

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 - boarding_pass
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 - flights
 - passengers
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 - receipt_airport
 - receipt_booking
 - receipt_booking_i
 - receipt_flight
 - receipt_passenge
 - security_check
 - sheep_counter
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sheepfarm/postgres@PostgreSQL 18*

Query

```
1 UPDATE baggage_check
2 SET check_result = 'Checked',
3     updated_at = CURRENT_TIMESTAMP
4 WHERE created_at >= TIMESTAMP '2024-03-01 00:00:00'
5     AND created_at < TIMESTAMP '2024-04-01 00:00:00'
6     AND check_result = 'Not checked'
7 RETURNING *;
```

Query History

Scratch Pad

Data Output Messages Notifications

baggage_check_id	check_result	created_at	updated_at	booking_id	passenger_id
[PK] integer	character varying (50)	timestamp without time zone	timestamp without time zone	integer	integer

Активация Windows

Чтобы активировать Windows, перейдите в раздел "Параметры".

Successfully run. Total query runtime: 73 msec. 0 rows affected.

Servers > PostgreSQL 18 > Databases > sheepfarm > Schemas > public > Tables > booking

CRLF Ln 7, Col 13